

CS 122 Setup Check

Exam setup verification. Everything stays local on the machine.

This verifies five things:

- VS Code can render markdown
- VS Code can compile and run a Java file
- JUnit 4 jars can be added to the project classpath
- VS Code can run JUnit tests and show green checkmarks
- Respondus LockDown Browser is installed and ready

Step 1: Create a Local Folder

Create a new folder anywhere on the local machine, name it **setup-check**. Open it in VS Code: **File → Open Folder**, choose the folder. Click "Yes, I trust the authors" if prompted.

Step 2: Verify Markdown Preview

Inside the folder, create a file named **README.md**. Paste:

```
# Test Header  
This text is bold when previewed.
```

Save. Right-click **README.md** in the file explorer and choose **Open Preview** (or press Cmd/Ctrl + Shift + V). The header should render large and the word "bold" should appear bold.

Step 3: Verify Java Compile and Run

Inside the folder, create **SetupCheck.java** and paste:

```
public class SetupCheck {  
  
    public static int add(int a, int b) {  
        return a + b;  
    }  
  
    public static void main(String[] args) {  
        System.out.println("VS Code Java setup is working.");  
        System.out.println("Sample sum: " + add(2, 3));  
    }  
}
```

Save. Run it using either:

1. Click the **Run** link that appears above **main**

2. Click the dropdown next to the play button at the top right of the editor, choose **Run Java** or **Run Code**

The terminal panel should print:

```
VS Code Java setup is working.  
Sample sum: 5
```

At least one method should produce that output. If both **Run Java** and **Run Code** fail, see Troubleshooting.

Step 4: Add JUnit 4 to the Project Classpath

JUnit is not part of the JDK, so it has to be added to the project manually. The class uses **JUnit 4** to match the lab convention. Ignore any VS Code prompts that point to JUnit 5 / Jupiter — those are not what is needed here.

Download the two required jars

Click each link in a browser to download the jar:

1. **junit-4.13.2.jar** <https://repo1.maven.org/maven2/junit/junit/4.13.2/junit-4.13.2.jar>
2. **hamcrest-core-1.3.jar** (required by JUnit 4's assertion methods)
<https://repo1.maven.org/maven2/org/hamcrest/hamcrest-core/1.3/hamcrest-core-1.3.jar>

Save both jars to a permanent folder — not Downloads, not Desktop. Downloads gets cleaned out on most machines, and the classpath stores absolute paths. If those files disappear later, the classpath silently breaks. Recommend creating a **junit-libs** folder in the home directory or Documents, and dropping both jars there. The same jars get reused for the exam project.

Add the jars to this project's classpath

1. Open the Command Palette: **Cmd/Ctrl + Shift + P**
2. Type: **Java: Configure Classpath**
3. Select that command. A "Project Settings" panel opens.
4. In the left sidebar of that panel, click **Classpath**, then in the top tabs, click **Libraries**
5. Click **+ Add Library...**
6. Navigate to where the jars were saved
7. Select both **junit-4.13.2.jar** and **hamcrest-core-1.3.jar** (hold Cmd/Ctrl to select both)
8. Click **Open**

Both jar paths should now appear in the Libraries list.

Create the test file

Inside the **setup-check** folder, create **SetupCheckTest.java** and paste:

```
import org.junit.Test;
import static org.junit.Assert.*;

public class SetupCheckTest {

    @Test
    public void testAddBasic() {
        assertEquals(5, SetupCheck.add(2, 3));
    }

    @Test
    public void testAddZero() {
        assertEquals(7, SetupCheck.add(7, 0));
    }
}
```

Save.

Step 5: Run the JUnit Tests

Run Test links should appear above each `@Test` annotation (small green play arrow icons in the gutter). Click **Run Test** on either method.

Look at the TEST RESULTS panel. Two green checkmarks next to `testAddBasic()` and `testAddZero()` means JUnit is working and the setup is exam-ready.

About the red squiggles on the imports

After adding the jars, the imports `org.junit.Test` and `org.junit.Assert` may **still show red squiggles** with "package org.junit does not exist" errors, even though the tests run green.

This is a known VS Code stale-cache issue, not a broken setup. The Test Runner has the classpath, but the Java Language Server (the part doing syntax checking) hasn't picked up the jars yet.

The success criterion is green checkmarks in TEST RESULTS, not the absence of red squiggles. If the tests run green, JUnit is configured correctly.

To clear the squiggles (escalating fixes):

1. **Reload the window:** Cmd/Ctrl + Shift + P → "Developer: Reload Window". Usually clears it within 5 seconds.
2. **Clean the language server workspace:** Cmd/Ctrl + Shift + P → "Java: Clean Java Language Server Workspace" → click "Restart and delete". Wait about 10 seconds for re-indexing. This almost always works.
3. **Quit and reopen VS Code entirely.**

Step 6: LockDown Browser

The exam runs through Respondus LockDown Browser. Without it installed, the exam will not start.

Install

Download from this link: <https://download.respondus.com/lockdown/download.php?ID=222638733>

- **Mac:** open the .dmg, drag the app into Applications
- **Windows:** run the installer and follow the prompts

Launch from Brightspace on Exam Day

1. Log in to Brightspace
2. Navigate to: **Class → Activities → Quizzes**
3. Click the exam quiz
4. The "Quiz Requirements" page shows two items:
 - A link to download and install LockDown Browser (skip if installed in the step above)
 - A blue **Launch LockDown Browser** button — click this to start the exam

On Exam Day

The exam folder will be a separate project. To use the same JUnit jars there:

1. Open the exam folder in VS Code
2. Cmd/Ctrl + Shift + P → "Java: Configure Classpath" → Libraries tab → Add Library
3. Select the same two jars from the permanent folder where they were saved

About 30 seconds. The jars are already downloaded.

Troubleshooting

No Run / Run Test links appearing, both Run Java and Run Code fail

- Open Extensions sidebar (Cmd/Ctrl + Shift + X)
- Search "Extension Pack for Java" by Microsoft
- Confirm installed AND enabled (the full bundle, not just one extension from it)
- Reload window: Cmd/Ctrl + Shift + P → "Developer: Reload Window"

Java not found

- Open terminal (View → Terminal)
- Run `java -version` and `javac -version`
- Both should return version numbers
- If not, the JDK is missing or not on PATH

Tests fail to RUN (not just squiggly red — actually fail to execute)

- Confirm BOTH jars were added (junit AND hamcrest); JUnit 4 needs both
- Confirm the imports are exactly `org.junit.Test` and `org.junit.Assert` (NOT `org.junit.jupiter.*` — that is JUnit 5 and will not work with these jars)
- Reload the window or clean the language server workspace (see Step 5)

Run Test link does not appear above @Test

- Confirm the test class is `public`

- Reload the window